MAT 2040: Linear Algebra

Assignment 6

- **Release date:** November 15, Friday.
- **Due date: November 24, Sunday.**
- Late submission is **Not** accepted.
- Please submit your answers as a PDF file with a name containing your student ID + ASS No. like "123456XXX ASS6.pdf".

1. Let $L : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear operator. If

$$L((1,2)^T) = (-2,3)^T \quad \text{and} \quad L((1,-1)^T) = (5,2)^T$$

find the value of $L((7,5)^T)$.

2. Determine whether the following are linear transformations from $\mathbb{R}^2$ to $\mathbb{R}^3$

   (a) $L(\mathbf{x}) = (x_1, x_2, 1)^T$

   (b) $L(\mathbf{x}) = (x_1, x_2, x_1 + 2x_2)^T$

   (c) $L(\mathbf{x}) = (x_1, 0, 0)^T$

   (d) $L(\mathbf{x}) = (x_1, x_2, x_1^2 + x_2^2)$

3. Determine whether the following are linear operators on $\mathbb{R}^{n\times n}$.

   (a) $L(A) = 2A$

   (b) $L(A) = A + I$

   (c) $L(A) = A - A^T$

4. Let T be a linear transformation that is both one-to-one and onto, then for each vector $\mathbf{w}$ in W there is a unique vector $\mathbf{v}$ in V such that $T(\mathbf{v}) = \mathbf{w}$. Prove that the inverse transformation $T^{-1} : W \to V$ defined by $T^{-1}(\mathbf{w}) = \mathbf{v}$ is linear.

5. Let L be a linear operator on a vector space V. Define $L^n, n \geqslant 1$, recursively by

$$L^1 = L, \quad L^{k+1}(\mathbf{v}) = L(L^k(\mathbf{v})) \quad \text{for all } \mathbf{v} \in V$$

Show that L^n is a linear operator on V for each $n \geqslant 1$.

6. Determine the kernel and range of each of the following linear operators on $\mathbb{R}^3$:

(a) $L(\mathbf{x}) = (x_3, x_2, x_1)^T$

(b) $L(\mathbf{x}) = (x_1, x_2, 0)^T$

(c) $L(\mathbf{x}) = (x_1, x_1, x_1)^T$

7. Let L be the linear operator on $\mathbb{R}^3$ defined by

$$L(\mathbf{x}) = \begin{bmatrix} 2x_1 - x_2 - x_3 \\ 2x_2 - x_1 - x_3 \\ 2x_3 - x_1 - x_2 \end{bmatrix}$$

Determine the standard matrix representation A is L, and use A to find $L(\mathbf{x})$ for each of the following vectors $\mathbf{x}$:

(a) $\mathbf{x} = (1, 1, 1)^T$

(b) $\mathbf{x} = (2, 1, 1)^T$

(c) $\mathbf{x} = (-5, 3, 2)^T$

8. Let

$$\boldsymbol{b}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \boldsymbol{b}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \boldsymbol{b}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

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and let L be the linear transformation from $\mathbb{R}^2$ into $\mathbb{R}^3$ defined by

$$L(\mathbf{x}) = x_1\boldsymbol{b}_1 + x_2\boldsymbol{b}_2 + (x_1 + x_2)\boldsymbol{b}_3$$

Find the matrix A representing L with respect to the ordered bases $\{\boldsymbol{e}_1, \boldsymbol{e}_2\}$ and $\{\boldsymbol{b}_1, \boldsymbol{b}_2, \boldsymbol{b}_3\}$.

9. Let L be the linear operator mapping $\mathbb{R}^3$ into $\mathbb{R}^3$ defined by $L(\mathbf{x}) = A\mathbf{x}$, where

$$A = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & -2 \\ 2 & -1 & -1 \end{bmatrix}$$

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and let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

Find the transition matrix V corresponding to a change of basis from $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ to $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$, and use it to determine the matrix B representing L with respect to $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

10. Find the standard matrix representation for each of the following linear operators:

    (a) L is the linear operator that rotates each $\mathbf{x}$ in $\mathbb{R}^2$ by 45° in the clockwise direction

    (b) L is the linear operator that reflects each vector $\mathbf{x}$ in $\mathbb{R}^2$ about the x_1 axis and then rotates it 90° in the counterclockwise direction.

    (c) L doubles the length of $\mathbf{x}$ and then rotates it 30° in the counterclockwise direction.

11. Suppose that $A = ST$, where S is nonsingular. Let $B = TS$. Show that B is similar to A.

12. Show that if A and B are similar matrices, then $\det(A) = \det(B)$.

13. Let A and B be similar matrices. Show that

    (a) A^T and B^T are similar.

    (b) A^k and B^k are similar for each positive integer k.

14. The trace of an $n \times n$ matrix A, denoted $\text{tr}(A)$, is the sum of its diagonal entries, that is,

$$\text{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}$$

    Show that

    (a) $\text{tr}(AB) = \text{tr}(BA)$

    (b) If A is similar to B, then $\text{tr}(A) = \text{tr}(B)$

15. Let A and B be similar matrices. Show that if λ is any scalar, then $\det(A - \lambda I) = \det(B - \lambda I)$.